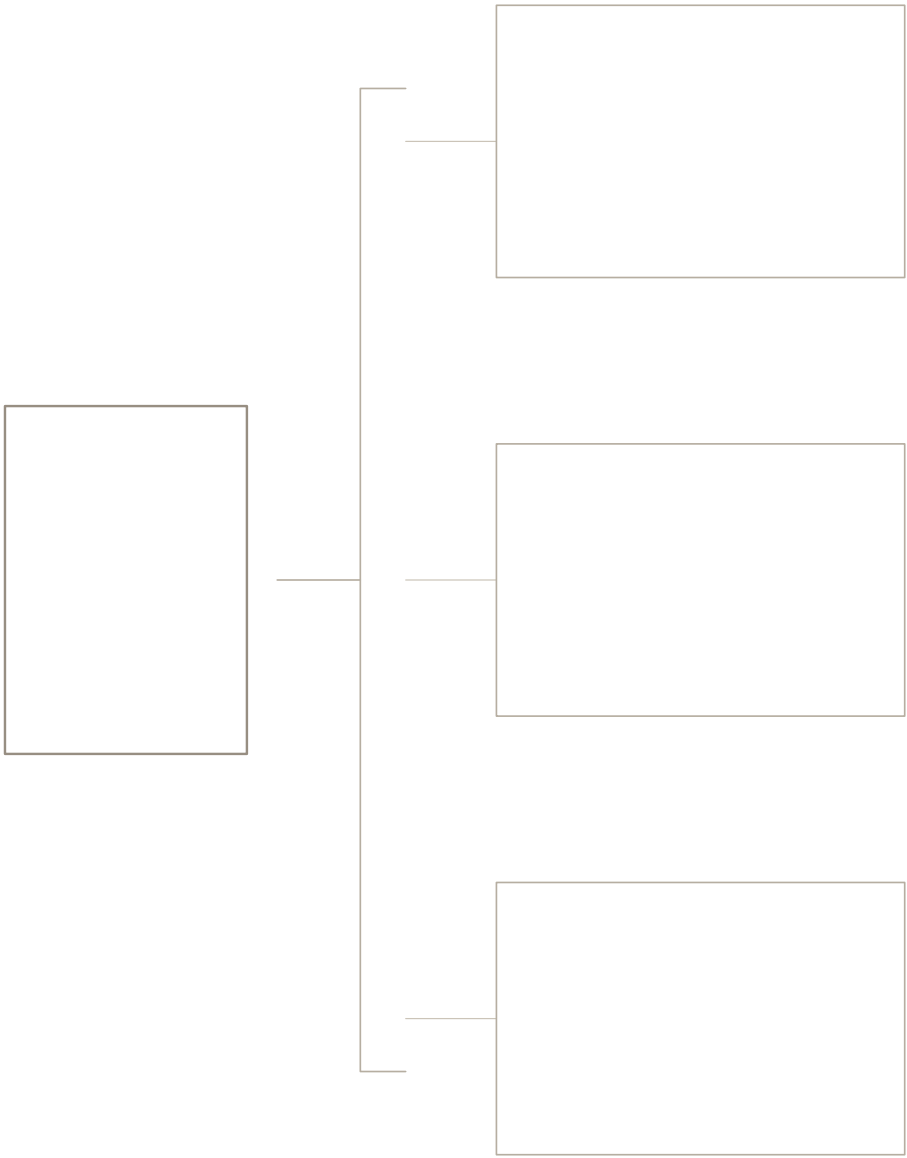Brace decomposition